Numerical notes for the accompanying code and data

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Phenotypic assortment and the evolution of cooperation in finite periodic phenotype spaces

Scope and notation

The archive reproduces seven figures for the analytical weak-selection threshold in the large-population mutation scaling. It does not simulate the finite-population Moran process. The threshold is denoted by $(b/c)_n^*(\alpha)$; its dependence on the parameters L, μ, and ν is left implicit. Here $L \geq 2$ and $n \geq 1$ are integers, $\alpha, \mu, \nu > 0$, $\mu = Nu/2$, and $\nu = Nv/2$.

Finite spectral expression

For $p = 0, \ldots, L-1$, define

$$c_p = \cos(2\pi p/L), \qquad \widehat{\varphi}(p) = \sum_{m=0}^{L-1} e^{-\alpha \min(m, L-m)} e^{2\pi i p m/L}.$$

The Fourier coefficients are real and positive for the kernel used here. The implementation uses the opposite Fourier sign, which gives the same coefficients because the kernel is even. For a mode $\mathbf{p} \in \{0, \ldots, L-1\}^n$, let

$$S_{\mathbf{p}} = n - \sum_{j=1}^{n} c_{p_j}, \qquad w_{\mathbf{p}} = L^{-n} \prod_{j=1}^{n} \widehat{\varphi}(p_j), \qquad \lambda_{\mathbf{p}} = \frac{2\nu}{n} S_{\mathbf{p}}.$$

These weights sum to one. The generalized Laplace transform is the finite sum

$$\mathcal{L}_n^{(L)}(\alpha, x) = \sum_{\mathbf{p}} \frac{w_{\mathbf{p}}}{S_{\mathbf{p}} + nx}, \qquad x > 0. \tag{1}$$

An equivalent form of the threshold avoids subtracting nearly equal Laplace transforms:

$$(b/c)_n^*(\alpha) = 1 + (1 + 2\mu)\frac{U}{V}, \tag{2}$$

where

$$U = \sum_{\mathbf{p}} \frac{w_{\mathbf{p}}}{(\lambda_{\mathbf{p}} + 1)(\lambda_{\mathbf{p}} + 2\mu + 1)}, \tag{3}$$

$$V = \sum_{\mathbf{p}} \frac{w_{\mathbf{p}} \lambda_{\mathbf{p}} (\lambda_{\mathbf{p}} + 2\mu + 2)}{(\lambda_{\mathbf{p}} + 1)(\lambda_{\mathbf{p}} + 2\mu + 1)(\lambda_{\mathbf{p}} + 2\mu + 3)}. \tag{4}$$

All summands are nonnegative and $V > 0$. The zero mode contributes zero to V.

The code first combines the modes p and $L-p$, then combines coordinate permutations with the multinomial factor $n!/\prod_j k_j!$. This leaves $\binom{n+\lfloor L/2 \rfloor}{n}$ terms. No mode is discarded. Direct grouped summation is used when this count is at most 200 000. All plotted thresholds use this method except $L = 100$, $n = 4, \ldots, 10$ in `fig3`.

Large spectra and the truncation bound

The remaining seven values are computed from the same finite spectrum using a convergent series. This avoids enumerating, for example, $\binom{60}{10} = 75\,394\,027\,566$ grouped terms for $L = 100$, $n = 10$.

Regard $Z = 1 - S_{\mathbf{p}}/(2n) \in [0, 1]$ as a random variable with probabilities $w_{\mathbf{p}}$, and define

$$M_k = \mathbb{E}[Z^k], \qquad J_k = \mathbb{E}[(1 - Z)Z^k].$$

The code computes both by finite binomial sums over the independent one-coordinate spectra. In particular, J_k is evaluated directly, not by subtracting M_{k+1} from M_k. Equation (1) also has the series representation

$$\mathcal{L}_n^{(L)}(\alpha, x) = \frac{1}{n(x+2)} \sum_{k=0}^{\infty} \left(\frac{2}{x+2}\right)^k M_k.$$

Its remainder after $k = K$ is at most $[2/(x+2)]^{K+1}/(nx)$.

For the threshold, write $B = 4\nu$ and $a = 1 + 2\mu$, so that $\lambda = B(1 - Z)$. For $s \in \{1, a, a+2\}$ define $d_s = B + s$ and $q_s = B/d_s$. Let

$$A_k = \frac{\sum_{i=0}^{k} q_1^i q_a^{k-i}}{d_1 d_a}, \qquad C_k = \frac{\sum_{i=0}^{k} q_1^i q_{a+2}^{k-i}}{d_1 d_{a+2}}.$$

Expanding the reciprocal factors in Eqs. (3)–(4) gives

$$U = \sum_{k=0}^{\infty} A_k M_k, \qquad V = \frac{B}{2} \sum_{k=0}^{\infty} (A_k + C_k) J_k.$$

Let U_K, V_K be the sums through $k = K$. Since $q_a, q_{a+2} \le q_1 < 1$, their nonnegative remainders satisfy

$$\begin{aligned} 0 \le U - U_K \le E_U &= \frac{T_K}{d_1 d_a}, \\ 0 \le V - V_K \le E_V &= \frac{B}{2}\left(\frac{1}{d_1 d_a} + \frac{1}{d_1 d_{a+2}}\right) T_K, \\ T_K &= \frac{q_1^{K+1}\big[(K+2) - (K+1)q_1\big]}{(1 - q_1)^2}. \end{aligned}$$

Thus the exact threshold lies in the interval

$$1 + a\frac{U_K}{V_K + E_V} \;\le\; (b/c)_n^*(\alpha) \;\le\; 1 + a\frac{U_K + E_U}{V_K}.$$

The code accepts $1 + aU_K/V_K$ only when its maximum distance from these bounds, divided by that value, is at most 10^{-12}. It raises an error if the available series orders do not meet this tolerance. The saved bounds cover series truncation; they do not include floating-point rounding.

Checks and saved data

The notebook runs 18 comparisons: grouped versus ungrouped sums, the series versus direct sums, and independent adaptive integration, including all $L = 100$ dimensions plotted. It also checks the special values $\mathcal{L}_1^{(3)}(1, 1/2) = 1 + e^{-1}$ and $(b/c)_1^*(1) \approx 12.45639491347548$ at $L = 3$, $\mu = 1/2$, $\nu = 1$. Quadrature is used only in these checks, never to generate a figure.

CSV files contain all values used in the plots; the two heat maps also have NumPy arrays. Their colors saturate at 10, but the stored values and contour calculations use the raw thresholds. See `DATA_DICTIONARY.txt` for grids and columns. The script `verify_results.py` compares new results with the supplied data using relative tolerance 5×10^{-11} and absolute tolerance 5×10^{-13}.